Diskrete Mathematik

Woche 5 - HG G 26.5

https://github.com/tschai-yim/eth-resources

Feedback (anonym): https://forms.gle/xnkcjsdj86LK6yMh7



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Heutige Agenda

1. Teil (45 min)
- 30 min: Theorie-Übersicht mit Beispielen
- 15 min: Übungen

10 min Pause

2. Teil (45 min): Übungen

Fragen zu alten Übungen?

→ in the Pause

Theorie-Übersicht

Kapitel 2.4, 2.6, 3.1-3.2

Predicate Logic: Syntax (recap)

- **Universe U: Non-empty set** of objects to reason about (e.g., $\mathbb{N}$, $\mathbb{R}$).
- **Constants:** Specific, fixed elements from U (e.g., $0, 1, \pi$).
- **Predicates $P(x)$:** Functions mapping elements from U to a truth value, $\{0, 1\}$.
  - Example: `less(x, y)` returns 1 iff $x < y$.
- **Functions $f(x)$:** Map elements from U to another element in U.
  - Example: `add(x, y)` on $\mathbb{N}$ returns $x + y$.
- **Variables:** Placeholders for elements from U (e.g., x, y, z).
- **Quantifiers:** $\forall x(\ldots)$ and $\exists x(\ldots)$

How to define $P(x)$ and $f(x)$?

$P(x) = 1 \Leftrightarrow x$ satisfies something

$f : A \to B, x \mapsto x + 1$

Proof Patterns (recap)

- Templates to prove things.
- **Do not memorize by heart!** Intuition comes naturally with practice.
- First major application of propositional logic.

Example indirect proof / contrapositive:

Lem 2.6. $\neg B \to \neg A \vDash A \to B$

Proofs for Implications ($S \Rightarrow T$) (recap)

- **Direct Proof** (Def. 2.13): Logically derive T.
- **Indirect Proof / Contrapositive** (Def. 2.14):
  1. **Assume** conclusion T is false ($\neg T$).
  2. Derive that premise S is false ($\neg S$).
- **Composition of Implications / Transitivity** (Def. 2.12):
  - Chain implications: $S \Rightarrow T \Rightarrow U$

> **Lem 2.5.** $(A \to B) \land (B \to C) \vDash A \to C$

Proofs for General Statements (S) (recap)

- **Modus Ponens** (Def. 2.15): Prove helper R, then prove $R \Rightarrow S$ to conclude S.
- **Case Distinction** (Def. 2.16): Show $R_1 \vee \ldots \vee R_k$, then prove $R_i \Rightarrow S$ for each i.
  - **Example:** For any real number x, $|x| \geq x$.
- **Contradiction** (Def. 2.17): Assume $\neg S$, derive a known falsehood T to force S.
  - **Example:** There is no largest natural number.

Proofs involving Quantifiers (recap)

- **Existence Proofs** (Def. 2.18): Prove $\exists x P(x)$
  - **Constructive:** Find concrete a where $P(a)$ holds.
  - **Non-constructive:** Prove an example must exist without identifying it.
    - **Example:** See pigeonhole example below.
- **Pigeonhole Principle** (Thm. 2.10)
  - n objects into $k < n$ bins guarantees one bin has $\geq \left\lceil \frac{n}{k} \right\rceil$ objects.
  - **Example:** In a room of 367 people, two must share a birthday.
- **Proof by Counterexample** (Def. 2.19): Disprove $\forall x P(x)$
  - **Constructive existence proof** for negation $\neg P(a)$.
  - **Example:** $x = \frac{1}{2}$ disproves $\forall x \in \mathbb{R}(x^2 \geq x)$, since $\left(\frac{1}{2}\right)^2 = \frac{1}{4} < \frac{1}{2}$.

Proof by Induction ($\forall n \in \mathbb{N} : P(n)$) (recap)

- **Principle:** Proves $\forall n \in \mathbb{N} : P(n)$
  1. **Basis Step:** Prove the initial case holds (e.g., $P(0)$ is true).
  2. **Induction Hypothesis (IH):** Assume $P(n)$ holds for an arbitrary $n \in \mathbb{N}$.
  3. **Induction Step:** Prove that $P(n) \Rightarrow P(n+1)$ using the IH.
  4. **Conclusion:** Conclude that $P(n)$ holds for all $n \in \mathbb{N}$ (Thm. 2.11).
- **Analogy (Domino Effect):**
  - Knock down the first domino (basis).
  - Ensure every falling domino knocks down the next (step).
  - Result: all dominos fall (conclusion).

Set Theory

- **Formalization** like Propositional and Predicate Logic.
- Informality breaks things $\rightarrow$ **Russell's Paradox**.
- Foundation for **everything**.
  - Can build cool stuff from functions to Turing Machines!
- Theoretically starts **with 9 Axioms**.
  - Only applications are important for us.

Set Theory: Basic Concepts

- **Set and Element ($\in$):**
  - Universe consists only of **sets**; elements are also sets.
  - **Elementhood predicate**: $x \in A$ ("x is an element of A").
- **Equality ($=$):**
  - **Axiom:** sets are equal $\Leftrightarrow$ they have the same elements.
  - Order and duplicates do not matter, e.g., $\{1, 2, 1\} = \{2, 1\}$.
  - **Core Lemma 3.2:** $A = B \Leftrightarrow (A \subseteq B \land B \subseteq A)$.
- **Subsets ($A \subseteq B$):**
  - Every element of A is also an element of B.
  - **Transitivity:** If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. [Lem 3.3]
- **The Empty Set ($\varnothing$ or $\{\}$):**
  - The **unique** set containing no elements. (Existence from Axiom)
  - It is a subset of every set: $\forall A(\varnothing \subseteq A)$.
- **Example:** $C = \{\varnothing\}$

Constructing and Operating on Sets

- **Construction:** Theoretically must prove a set exists.
  - Axioms guarantee **singleton sets** $\{a\}$ and finite sets $\{a, b, c\}$.
- **Operations:**
  - **Union ($A \cup B$):** elements in A, or in B, or in both.
  - **Intersection ($A \cap B$):** elements in both A and B.
  - **Difference ($B \setminus A$):** elements in B but not in A.
- **Power Set ($P(A)$):** set of **all subsets** of A.
  - Cardinality: If $|A| = k$, then $|P(A)| = 2^k$.
- **Cartesian Product ($A \times B$):** ordered pairs (a, b) with $a \in A$ and $b \in B$.
  - Cardinality: $|A \times B| = |A| \cdot |B|$.
  - Pair Definition: $(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}$, ensuring $(a, b) = (c, d) \Leftrightarrow a = c \wedge b = d$.

Übungen

https://github.com/tschai-yim/eth-resources unter
`03_TA/26HS_252-0025-01L_Diskrete_Mathematik/exercises`

Offene Fragen & Feedback

- Gibt es noch Fragen zu den heutigen Themen oder den Übungen?
- Habt ihr Feedback zur heutigen Übungsstunde?
  - Anonymes Formular: https://forms.gle/xnkcjsdj86LK6yMh7
- Bei Fragen: Schreibt mir eine E-Mail oder auf WhatsApp.

Schöne Woche und bis nächsten Montag!